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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Experiment – 7: Database as a Service(DBaaS)

Aim: To create and Connect to a MySQL Database with Amazon RDS. (Free Tier Cloud Platform).

Objective: After performing the experiment, the students will be able to learn –
- Create an environment to run MySQL database
- Connect to the database
- How To Access RDS Database
- Delete the database instance

Lab objective mapped : ITDO8024.3 : To learn and examine cloud service model and deploy database, storage services over different cloud platforms.

Prerequisite:
An AWS account,
MySQL Workbench application.

Requirements: Cloud Login**,** Desktop, Browser, Internet, GitHub etc.

Pre-Experiment Theory: (describe following topics in soft copy form)

1. **Database services-**

Sr No	Database type		AWS service
1	Relational Database	Relational databases store data with predefined schemas and relationships between them. These databases are designed to support ACID transactions, and maintain referential integrity and strong data consistency.	1. Amazon Aurora 2. Amazon RDS 3. Amazon Redshift
2	In-memory Database	In-memory databases are used for applications that require real-time	1. Amazon ElastiCache

		access to data. By storing data directly in memory, these databases deliver microsecond latency to applications for whom millisecond latency is not enough.	2. Amazon MemoryDB for Redis
3	Document Database	A document database is designed to store semistructured data as JSON-like documents. These databases help developers build and update applications quickly.	Amazon DocumentDB (with MongoDB compatibility)
4	Key-value Database	Key-value databases are optimized for common access patterns, typically to store and retrieve large volumes of data. These databases deliver quick response times, even in extreme volumes of concurrent requests.	Amazon DynamoDB

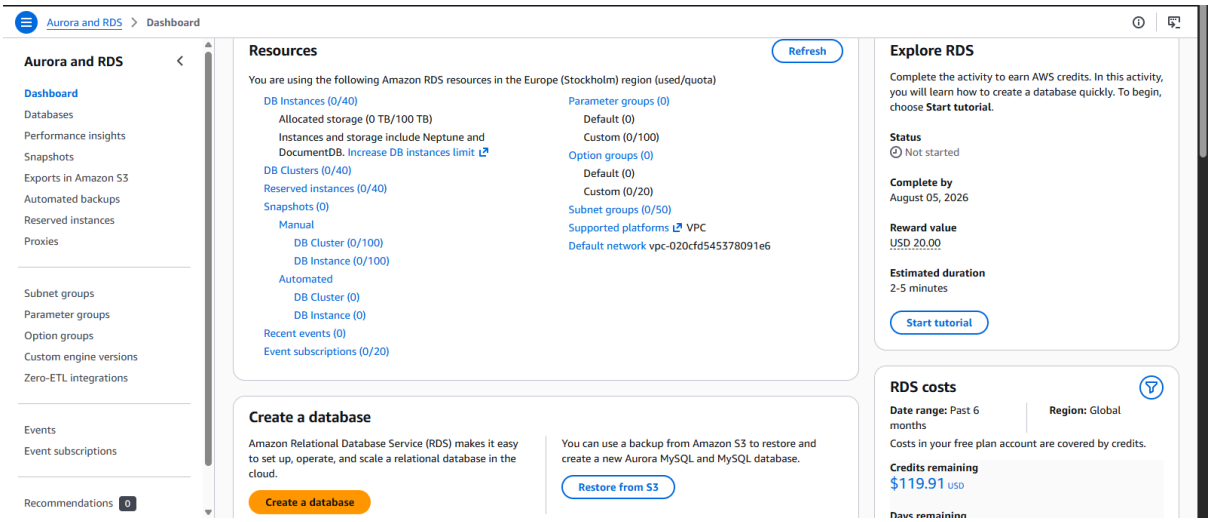
2. **Amazon RDS**

Amazon RDS is an easy to manage relational database service optimized for total cost of ownership. It is simple to set up, operate, and scale with demand. Amazon RDS automates the undifferentiated database management tasks, such as provisioning, configuring, backups, and patching. Amazon RDS enables customers to create a new database in minutes, and offers flexibility to customize databases to meet their needs across 8 engines and 2 deployment options. Customers can optimize performance with features, like Multi-AZ with two readable standbys, Optimized Writes and Reads, and AWS Graviton3-based instances, and choose from multiple pricing options to effectively manage costs.

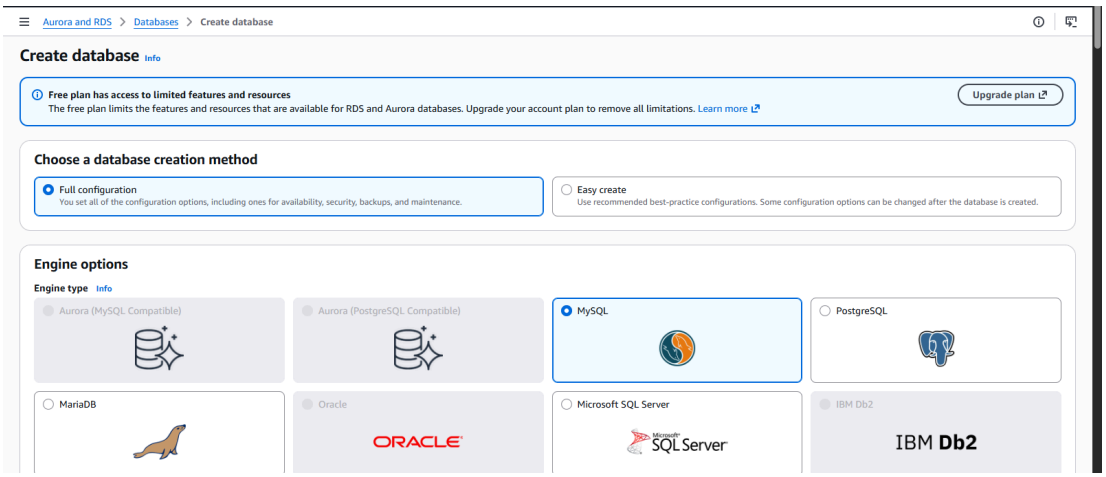
Fig. Amazon Relational Database Service (Amazon RDS) working

Procedure

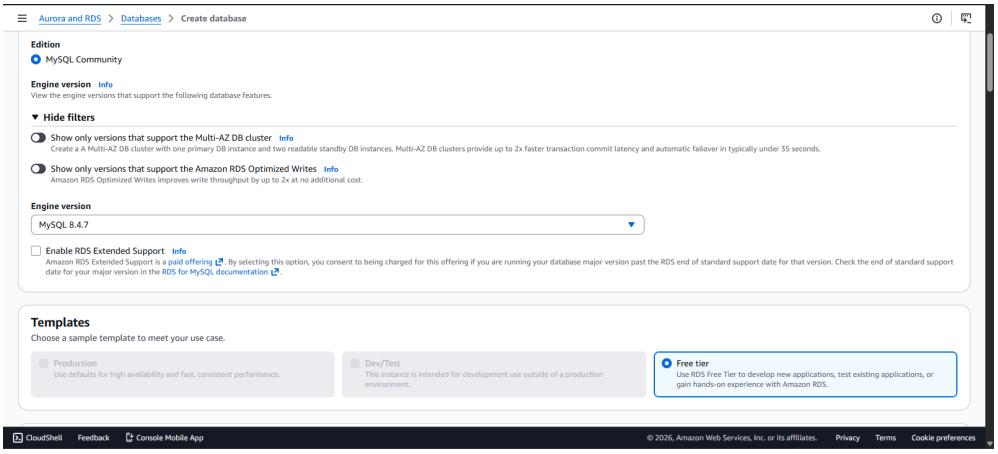
Step1: Create MY SQL DB instance



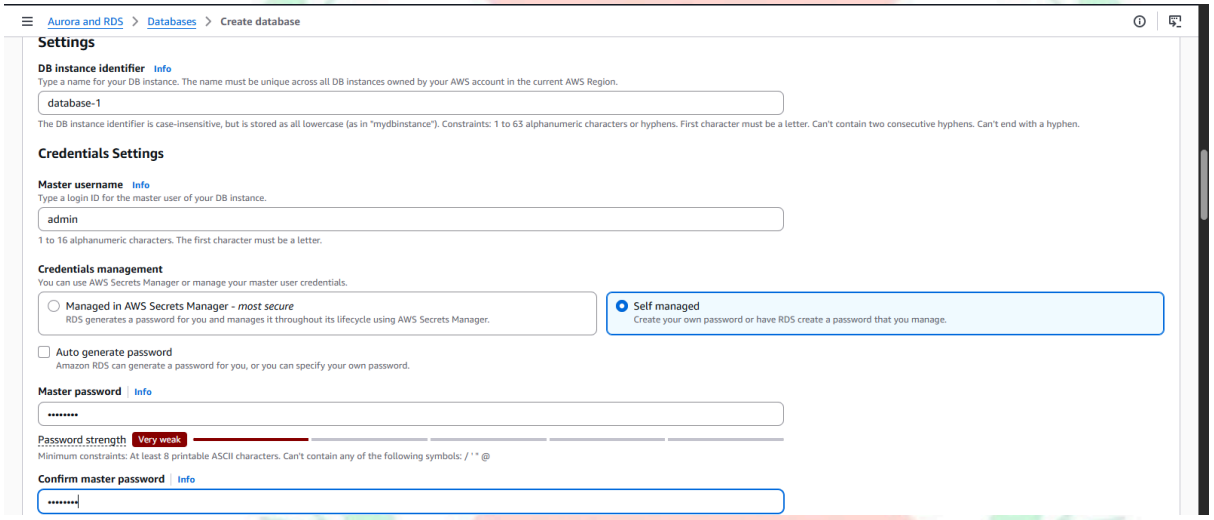
1. select AWS RDS service
2. change to N.vergina region
3. select option ‘create database’
4. select option ‘standard create’ and not ‘easy create’
5. select engine option as MYSQL



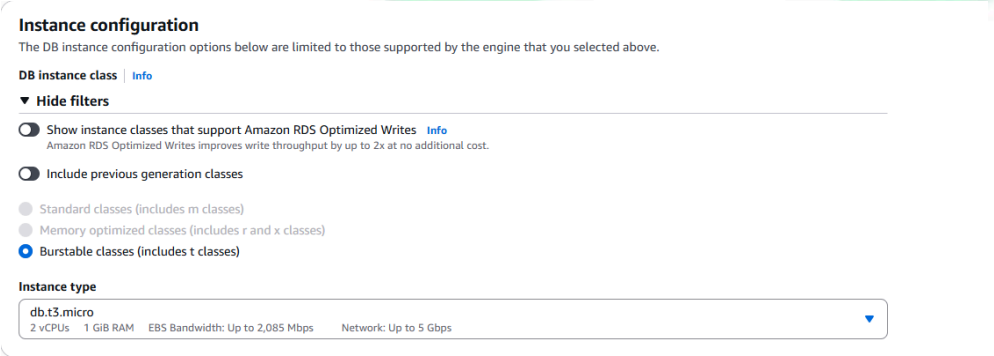
edition- MYSQL community
engine version -MYSQL (some version number)
Template: free tier
availability and durability - all disable



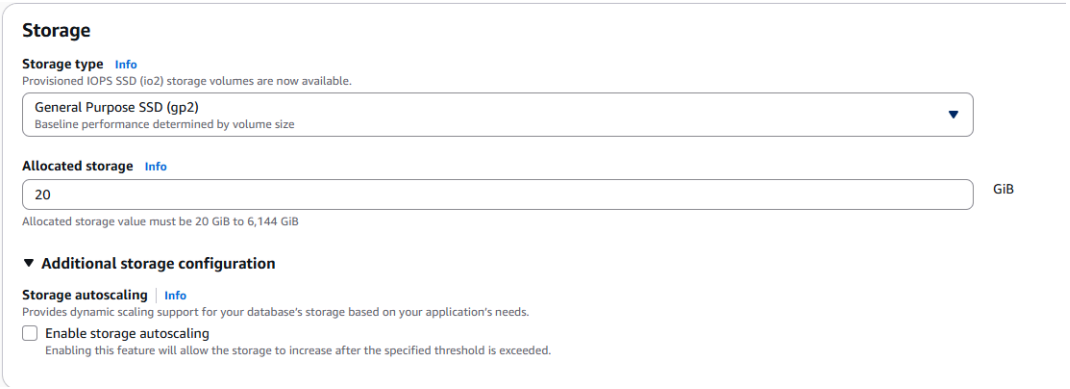
6. DB instance configuration steps:
   settings:
   DB instance identifier - Name DB
   Master uname:- admin
   Master pwd: (give some pwd)
   confirm pwd: (retype)



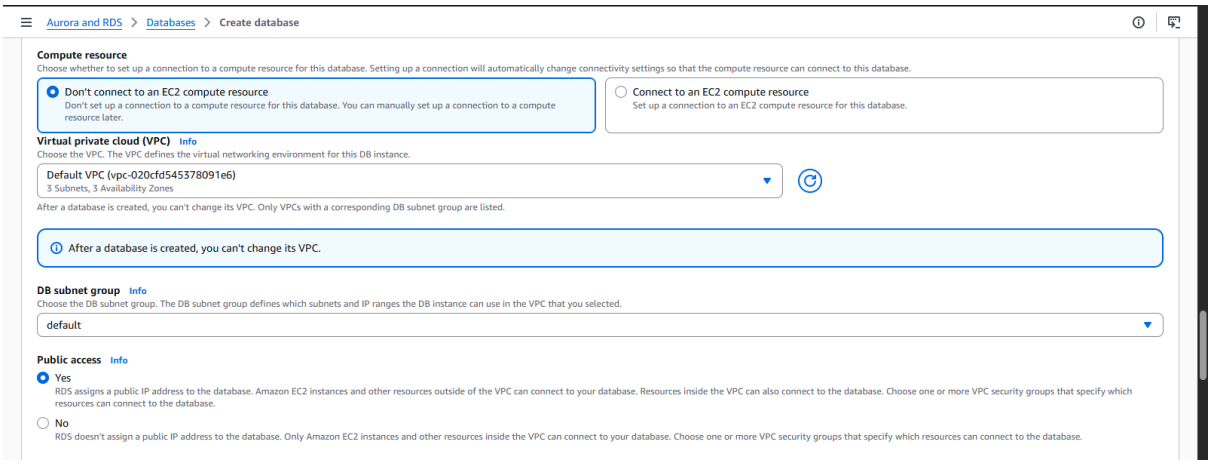
7. Instance configuration:
   DB instance class: Burstable classes
   select db.t2.micro



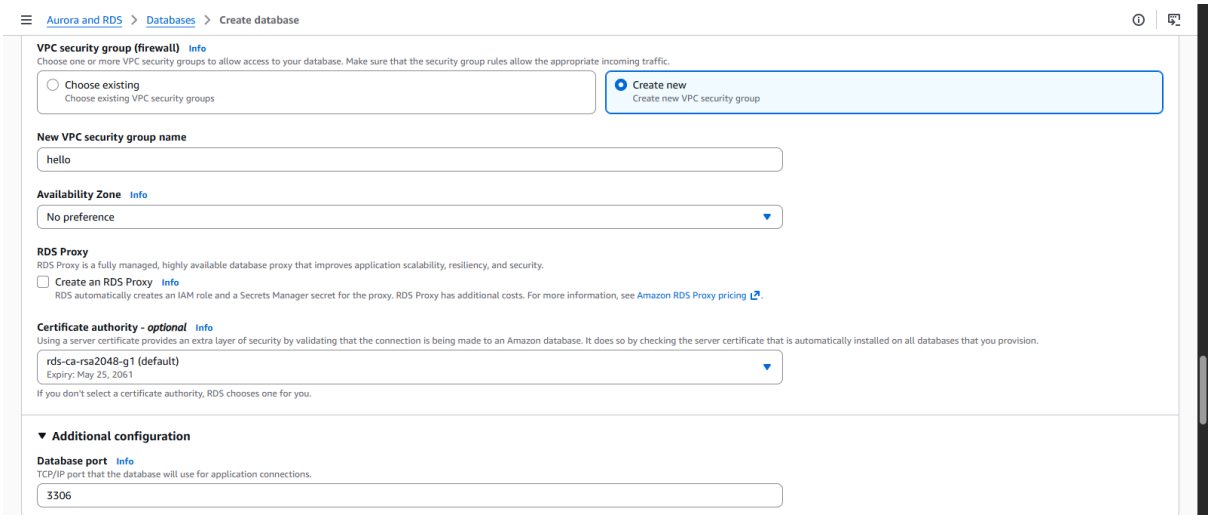
8. Instance specifications:
   storage type: General purpose (SSD)
   allocated storage- 20 GB
   enable storage autoscaling - disable



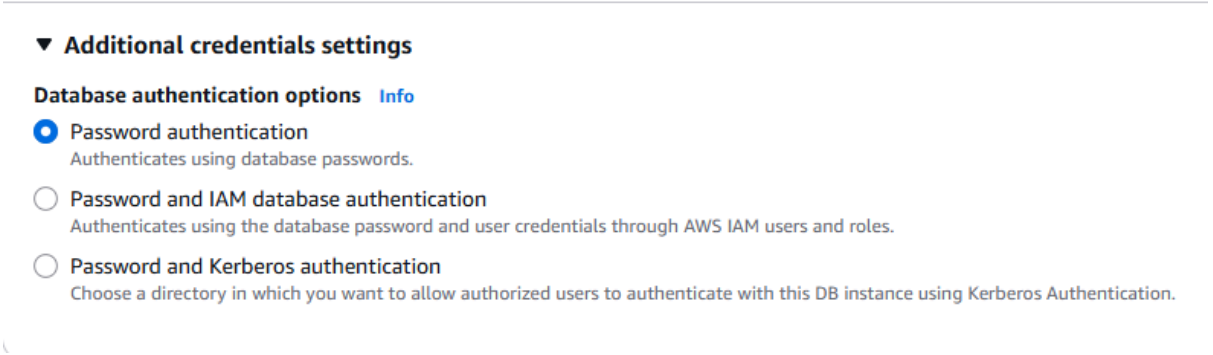
9. connectivity
   compute resource: -Don't connect toan EC2 compute resource
   VPC -default
   subnet - default
10. Public accessibility: yes RDS assignes a public IP address.
    (This setting will enable direct connection to the database from your device)



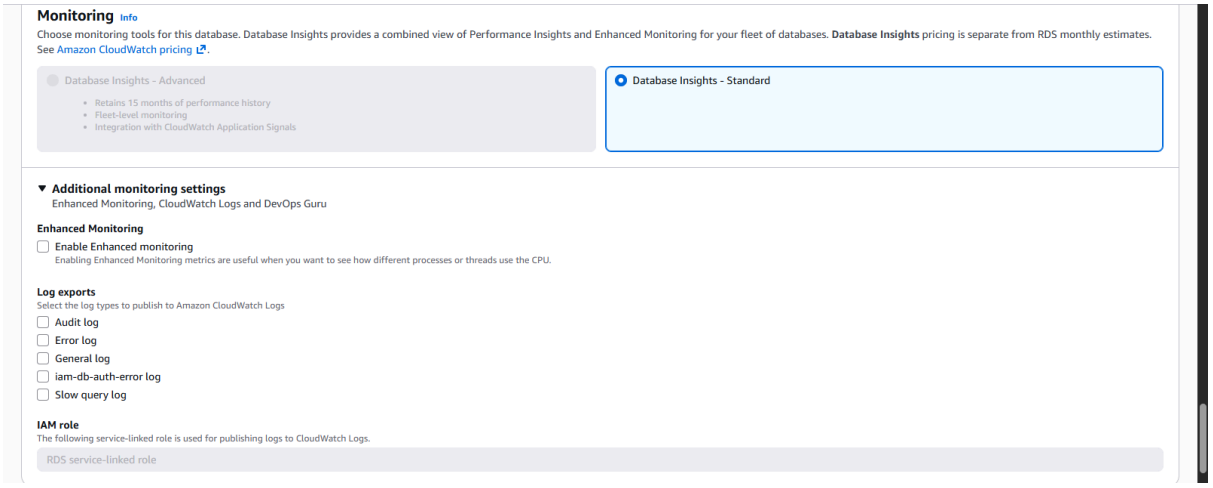
11. VPC security group:
    create new VPC security group (this will allow connection from the IP address of the device to created database)
    Name new security group
12. availability zone: No preference
13. RDS proxy - unchecked
14. Additional configuration:
    Port -3306 (default)



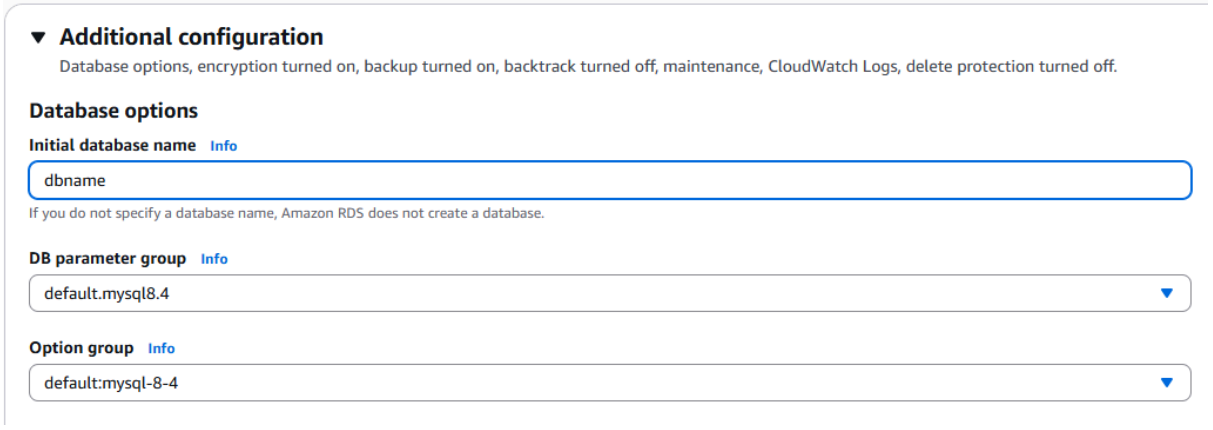
15. Database Authentication: select pwd authentication



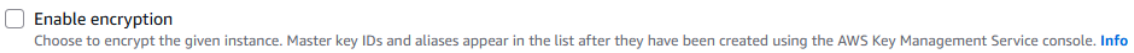
16. Monitoring:
    Enhanced monitoring unchecked [paid service- allow to check metrics in real time for the os that DB instance runs on]



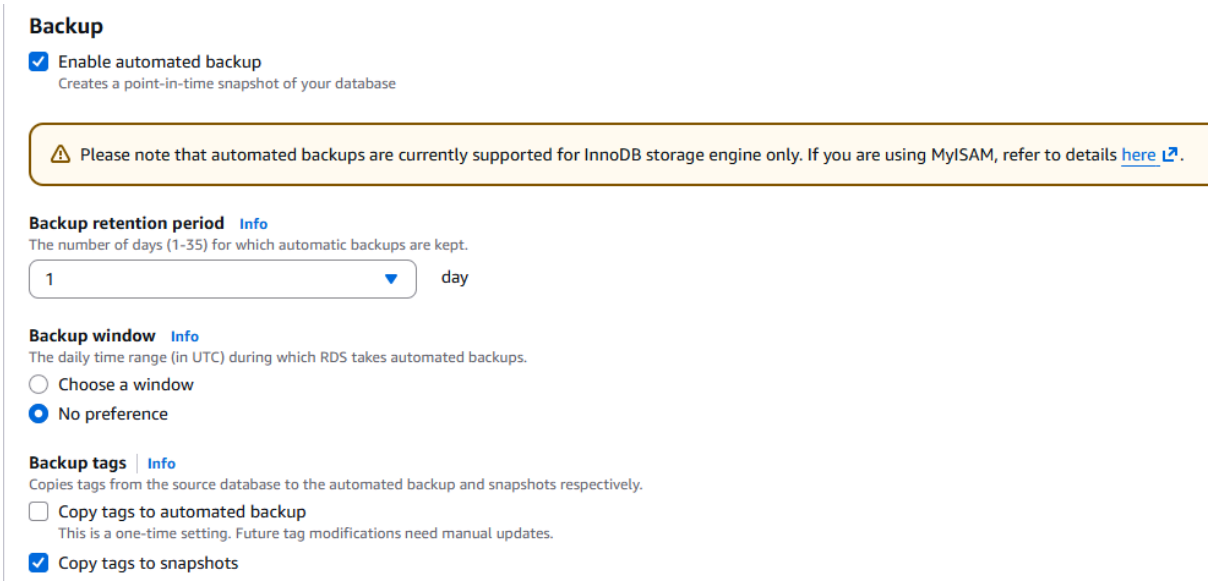
17. Additional Configuration:
    Database options
    name- dbname(compulsary) AWS RDS does not create it automatically if not specified.
    DB parameter group - default
    option group - default



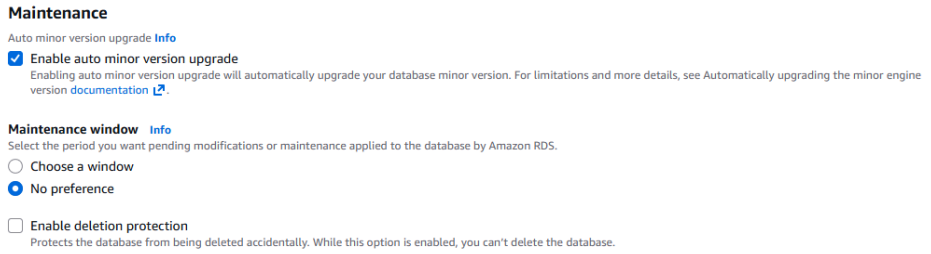
Encryption -not for free



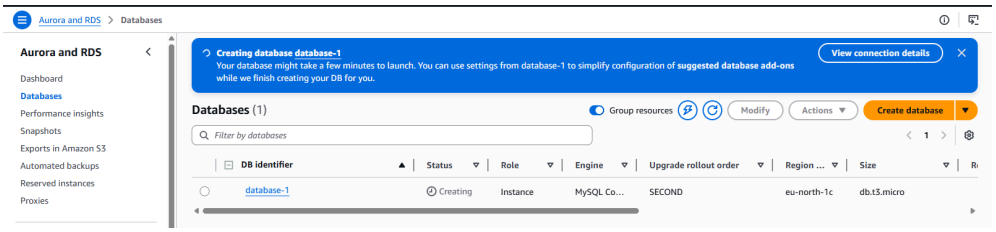
Back up - retention period 1 day
Back up window- no preference
copy tag to snapshot - checked

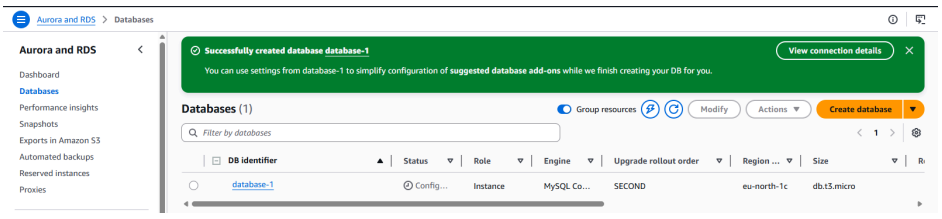


log exports - all unchecked
IAM role -default
Maintainance:
enable auto mirror - checked
maintenance window - no preference
Deletion protection: enable -unchecked



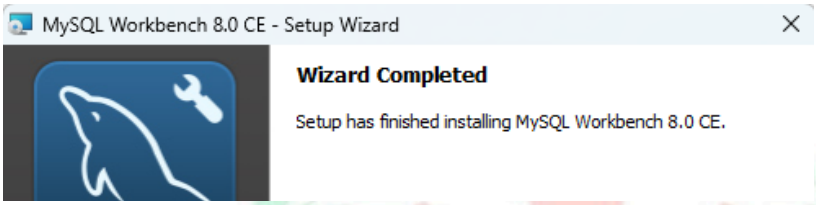
18. select Create Database
DB is being created.
Status changing to Creating to backing-up to Available (indicating DB is running)





Step2: Download SQL workbench client

Download MYSQL Workbench client and install . (No sign-up required)



Step3: Connect to MySQL DB using SQL workbench client

Launch MySQLWorkbench application.

Select Database - connect to database

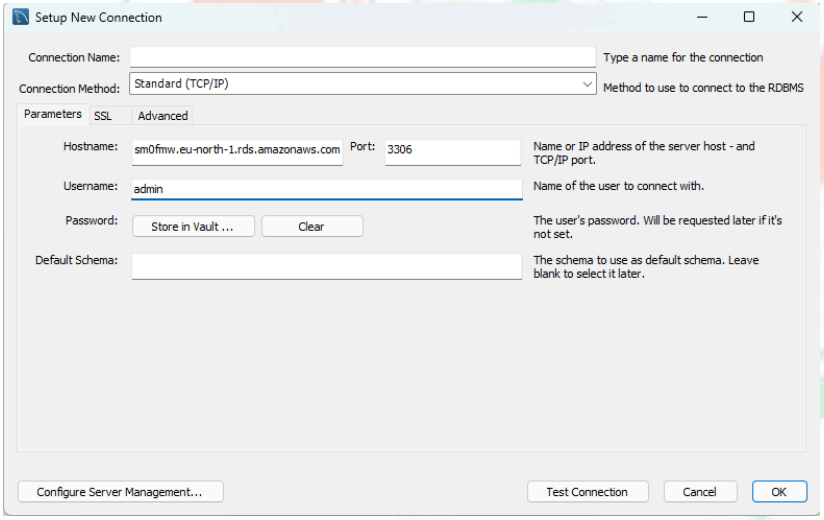
Enter the following:

Hostname : endpoint info of RDS

Port : 3306

Username and pwd: Use the ones set earlier while creating AWS RDS DB.

Pwd stored in vault - ok



Run the following SQL quires on SQL workbench client

To execute the SQL statement highlight the statement from the coding window and execute. Observe the action output.

```
CREATE  DATABASE ex;

USE ex;

CREATE TABLE st(
  table_id INT PRIMARY KEY,
  stud_id  INT,
  stud_name CHAR(10));

desc st;

INSERT INTO st( table_id, stud_id, Stud_name) values (01, 100, xyz)
```

SELECT * FROM st;
DELETE FROM st WHERE table=01;
DROP TABLE st;

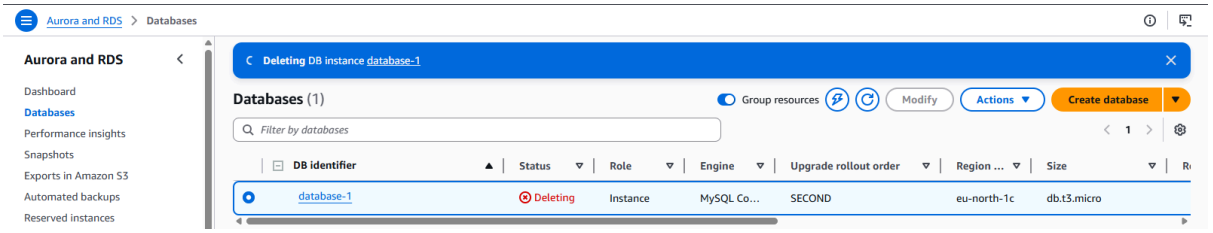
Output:

Output

Action Output

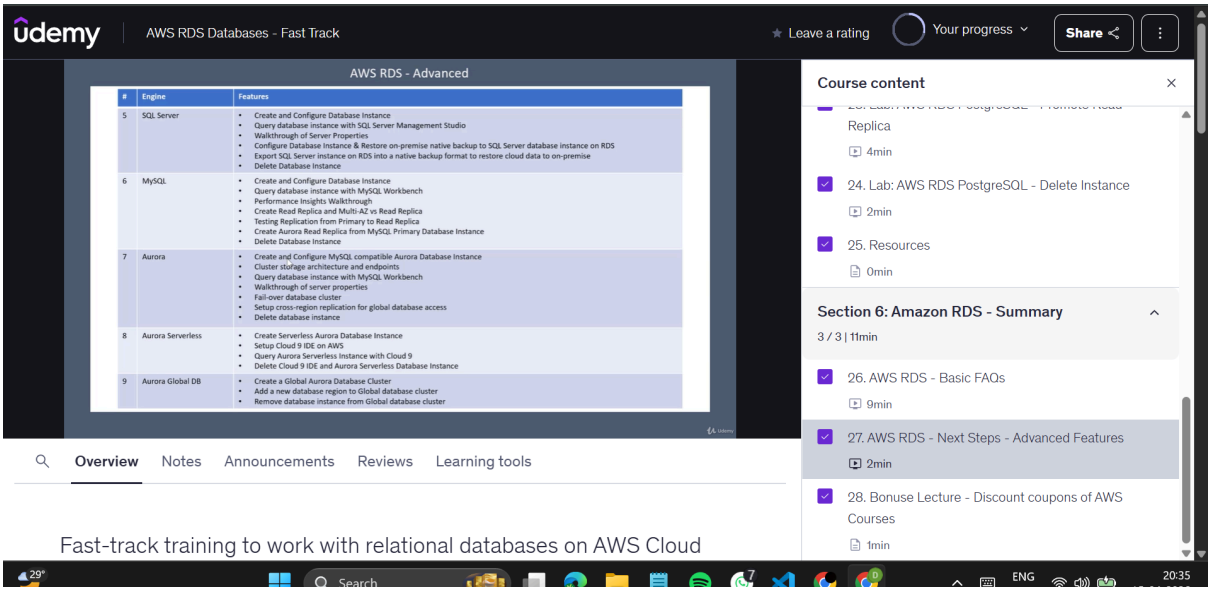
#	Time	Action	Message	Duration / Fetch
1	14:40:39	CREATE DATABASE ex	1 row(s) affected	0.172 sec
2	14:40:39	USE ex	0 row(s) affected	0.157 sec
3	14:40:39	CREATE TABLE st (table_id INT PRIMARY KEY, stud_id INT, stud_name CHAR(10))	0 row(s) affected	0.219 sec
4	14:40:39	DESCRIBE st	3 row(s) returned	0.171 sec / 0.000 sec
5	14:40:40	INSERT INTO st (table_id, stud_id, stud_name) VALUES (1, 100, 'xyz')	1 row(s) affected	0.172 sec
6	14:40:40	SELECT * FROM st LIMIT 0, 1000	1 row(s) returned	0.157 sec / 0.000 sec
7	14:40:40	DELETE FROM st WHERE table_id = 1	1 row(s) affected	0.156 sec
8	14:40:40	DROP TABLE st	0 row(s) affected	0.187 sec

Step 4: Clean up DB instance
Delete the DB instance and security group created inside AWS RDS.



Post-Experiments Exercise

AWS RDS Databases - Fast Track (free course on Udemy):
https://www.udemy.com/course/aws-rds-databases-tutorial-training-free/
complete the course and attach completion proof (like enrollemnet proof, prepare one google_quiz_10mrks_attach link here, proof of notes taken etc)



(no certificate provided for free enrollments)

Extended Theory:

1. Why are databases essential?(to be written in hand)
2. How many DB instances can I run with Amazon RDS?(to be written in hand)
3. Types of AWS Databases(Complete given table)(soft copy)

Database Providers	AWS	AZURE	GOOGLE
NAMES	1. Amazon Aurora	1. Azure SQL Database	1. Cloud SQL
	2. Amazon Redshift	2. Azure Cosmos DB	2. BigQuery
	3. Amazon RDS	3. Azure Database for MySQL/PostgreSQL	3. Cloud Spanner
	4. Amazon DynamoDB	4. Azure Synapse Analytics	4. Firestore

Results/Calculations/Observations:

Fill the following observation tables-(to be written in hand)

Amazon Aurora Vs RDS

Sr. No	**Parameter**	**Amazon Aurora**	**Amazon RDS**
1.	Performance		
2.	Compatibility with Database Engines		
3.	Limitations		

Post Experimental Exercise-(soft copy)

Questions:

1. **What is a database instance (DB instance)?**

ANS: A **DB Instance** is an isolated database environment in Amazon RDS.

2. It is where your **actual database runs** (like MySQL, PostgreSQL, etc.).
3. Includes **CPU, memory, storage, and networking resources**.
4. Managed by AWS, so **no need to handle hardware or maintenance manually**.
5. Supports **backup, recovery, and scaling features**.
6. Each DB instance has a **unique endpoint** to connect applications.
7. Can be configured with **security groups and access control**.

2. How will I be charged and billed for my use of Amazon RDS?

ANS:

- Billing is **pay-as-you-go** (you pay only for what you use).
- Charges are based on:

  - **DB Instance hours** → Time your instance is running
  - **Storage** → Amount of database storage used
  - **I/O requests** → Number of read/write operations
  - **Backup storage** → Automated backups and snapshots
  - **Data transfer** → Data sent out of AWS
  - **Multi-AZ deployment** → Extra cost for high availability
- Different pricing options:
  - **On-Demand** → Pay per hour (flexible)
  - **Reserved Instances** → Lower cost for long-term usage
  - **Free Tier** → Limited free usage for beginners
- Billing is shown in the **AWS Billing Dashboard**.

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:
[1][Online]https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Migrating.RDSMySQL.Import.html#AuroraMySQL.Migrating.RDSMySQL.Space
[2] [Online]https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Welcome.html